



iostat Command in Linux

Introduction

Imagine you need to:

- Monitor disk and CPU usage in real time.
- Find out if your disk is too slow or overloaded.
- Analyze system performance over time.

How do you track disk performance efficiently?

This is where the **iostat** (Input/Output Statistics) command helps!

Section 1: Learn

What is **iostat**?

The **iostat** command in Linux displays CPU and disk usage statistics. It helps:

1. Analyze disk performance and find bottlenecks.
2. Monitor CPU and I/O usage for performance tuning.
3. Identify slow or overloaded disks.

Why Do We Need **iostat**?

Without iostat	With iostat
No way to check disk performance	Displays real-time disk usage
Hard to find I/O bottlenecks	Helps diagnose slow disks
No CPU utilization insights	Provides CPU load details



Without iostat	With iostat
Need third-party tools for disk analysis	Built-in system monitoring tool

How Does **iostat** Work?

1. It collects CPU and disk usage statistics.
2. It updates the data periodically to show trends over time.
3. Users analyze the data to optimize system performance.

An Interesting Anecdote

In early Unix systems, **disk bottlenecks caused major slowdowns**, and system admins had no way to track them. The **iostat** command was introduced to **help diagnose slow I/O operations**, improving system performance!

Section 2: Practice

Step-by-Step Guide to Using **iostat**

1. Checking Basic CPU and Disk Statistics

```
# Display basic CPU and disk usage  
iostat
```

What happens here?

- Shows **CPU usage percentage**.
 - Displays **disk read/write speed**.
-



2. Displaying Human-Readable Output (-h)

```
# Show disk usage in human-readable format
```

```
iostat -h
```

Why is this useful?

- Makes **disk I/O stats** easier to understand.
-

3. Monitoring Disk Usage with Multiple Updates (-d 5)

```
# Show disk statistics every 5 seconds
```

```
iostat -d 5
```

What does this do?

- Helps **monitor disk activity** continuously.
-

4. Displaying CPU and Disk Usage Together (-c -d)

```
# Show CPU and disk stats side by side
```

```
iostat -c -d
```

Why is this important?

- Helps **compare CPU and disk performance** together.
-

5. Monitoring Specific Disk Devices (-p)

```
# Show statistics for a specific disk (e.g., sda)
```

```
iostat -p sda
```

How does this help?

- Useful when **analyzing disk-specific performance**.



6. Checking Performance Every 2 Seconds for 3 Updates (2 3)

```
# Show 3 updates at 2-second intervals
```

```
iostat 2 3
```

Why is this useful?

- Tracks **changes in disk and CPU usage over time.**

Real-World Example

A **system administrator** notices **slow database queries**. They need to:

1. **Check if the disk is overloaded.**
2. **Monitor I/O performance in real time.**
3. **Optimize database operations if needed.**

What did they do?

1. Ran **iostat -d 5** to **monitor disk activity continuously.**
2. Noticed high **disk read/write usage**, confirming an I/O bottleneck.
3. Optimized **database queries to reduce disk load.**

Result? The system **performed faster, and queries ran smoothly!**

Section 3: Know More

Frequently Asked Questions

1. **How do I install iostat if it's missing?**

```
sudo apt install sysstat # Ubuntu/Debian
```

```
sudo yum install sysstat # RHEL/CentOS
```



2. How do I see only CPU usage?

```
iostat -c
```

3. How do I monitor a specific disk only?

```
iostat -p sdb
```

4. Why is my disk utilization so high?

- Check which processes are using the disk:

```
sudo iotop
```

5. Where can I practice using **iostat**?

- Use a **Linux Virtual Machine**.
- Monitor disk usage on **cloud servers like AWS**.

Conclusion

The **iostat** command helps monitor disk and CPU usage, diagnose performance issues, and optimize system efficiency.

Start by checking basic disk stats, monitoring usage over time, and analyzing specific devices, and soon you'll be managing Linux performance like a pro!